

Let's understand what each query implies for the business

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1) Fetch all the paintings that are not displayed in any museums?

Insight / Finding:

The analysis identifies paintings that currently do not have any museum allocation. These artworks exist in the collection database but are not being displayed or associated with any museum, which may indicate underutilized assets or incomplete collection mapping.

Possible Reasons / Investigation Areas:

- **Temporary storage:** Paintings may be kept in storage due to limited exhibition space or rotation policies.
- **Renovation or maintenance:** Artworks may not be displayed because the assigned museum is under renovation or the artwork is undergoing restoration.
- **Collection planning decisions:** Some paintings may intentionally not be displayed due to curatorial choices, historical importance, or exhibition scheduling.
- **Data quality issue:** Museum assignment details may be missing or incorrectly recorded in the database.
- **Newly acquired artwork:** Recent acquisitions may not yet have been assigned to a museum.

Business Impact / Recommendation:

Museums should review these unassigned paintings to improve collection utilization. The management team can evaluate whether these artworks should be included in upcoming exhibitions, transferred to suitable museums, restored, or updated in the database to maintain accurate inventory records.

2) Are there museums without any paintings?

Insight / Finding:

The analysis identifies museums that currently do not have any paintings associated with them. These museums exist in the database but have no artwork allocation, which may indicate inactive museums, underutilized facilities, incomplete collection records, or missing data relationships.

Possible Reasons / Investigation Areas:

Museum closure: Museums may have been permanently closed but still remain in the database without updated records.

Renovation or maintenance: Museums may temporarily not have paintings because they are under renovation or collections have been moved elsewhere.

Low visitor demand or location challenges: Museums in less accessible locations may have limited collections or reduced operational activity.

Data quality issue: Paintings may exist but are not correctly mapped to the respective museum due to missing or incorrect database entries.

Newly established museum: Newly opened museums may not have acquired or uploaded their painting collections yet.

Business Impact / Recommendation:

Museums with zero paintings should be further investigated to understand whether the issue is operational or related to data management. The management team should verify museum status, improve artwork allocation records, and ensure collections are effectively utilized to support exhibitions, visitor engagement, and accurate inventory tracking.

3) How many paintings have an asking price of more than their regular price?

Regular Price = normal/base price of the painting

Asking Price = the price currently being requested from buyers/collectors

Insight / Finding:

The analysis identifies paintings where the current asking price is higher than the regular price, indicating artworks that are being positioned at a premium value. These paintings may represent high-demand, rare, or strategically priced assets within the collection.

Possible Reasons / Investigation Areas:

Higher market demand: Multiple collectors or buyers may show interest, leading sellers to increase the asking price.

Auction or premium selling strategy: Some artworks may be priced higher due to competitive bidding, limited availability, or collector interest.

Artist reputation or artwork uniqueness: Paintings created by renowned artists or having historical significance may justify higher pricing.

Incorrect regular pricing: The original regular price may be outdated and not reflect the current market value.

Pricing strategy: Sellers may intentionally set a higher asking price, expecting negotiation or future appreciation.

Business Impact / Recommendation:

These paintings should be analyzed further to understand whether the higher asking price is driven by genuine market value or pricing inconsistencies. Organizations can review artwork valuation, demand trends, and pricing strategies to maximize revenue and maintain accurate asset valuation.

4) Identify the paintings whose asking price is less than 50% of their regular price

Insight / Finding:

The analysis identifies paintings that are being offered at a significantly lower asking price compared to their regular price. These artworks may indicate discounted inventory, reduced market demand, or a mismatch between the original valuation and current market expectations.

Possible Reasons / Investigation Areas:

Low market demand:

The paintings may have limited buyer interest, causing sellers to reduce prices to attract customers.

Artist popularity or recognition:

Artworks created by less-known artists may have lower demand, resulting in lower selling prices.

Long inventory holding period:

Paintings that have remained unsold for a long time may be discounted to clear inventory and recover value.

Artwork condition issues:

Some paintings may have damage, restoration requirements, or quality concerns affecting their market value.

Incorrect initial valuation:

The regular price may have been set too high compared to the actual market value of the artwork.

Change in market trends:

Collector preferences may shift over time, reducing demand for certain styles or artists.

Business Impact / Recommendation:

Museums or art sellers should analyze these discounted paintings to understand whether price reductions are driven by genuine market conditions or valuation issues. They can review artwork popularity, condition, historical sales trends, and customer interest to improve pricing strategies and avoid undervaluing valuable assets.

5) Which Canva size costs the most?**Insight / Finding:**

The analysis identifies the canvas size with the highest cost among available canvas options. This helps understand which canvas dimensions are associated with higher pricing and may indicate premium canvas categories within the art supply market.

Possible Reasons / Investigation Areas:**Higher material usage:**

Larger canvas sizes may require more raw materials, leading to increased production costs.

Better quality or premium materials:

The expensive canvas may use higher-quality fabric, coating, durability features, or professional-grade materials preferred by artists.

Artist preference and demand:

Certain canvas sizes may be widely preferred by professional artists, increasing demand and market value.

Professional artwork usage:

Some canvas dimensions may be commonly used for exhibitions, galleries, or large-scale artworks, creating higher demand.

Brand reputation or marketing impact:

Popular canvas brands or heavily marketed sizes may command higher prices due to customer trust and recognition.

Supply availability:

Limited availability or production challenges for certain canvas sizes may increase their market price.

Business Impact / Recommendation:

Art sellers and suppliers should analyze whether the higher canvas cost is driven by genuine factors such as material quality, artist demand, or production cost. Understanding these factors can help optimize pricing strategies, maintain competitive pricing, and ensure premium canvas products are positioned correctly in the market.

Additional Business Recommendation:

The business can explore **bundling strategies** by combining premium canvas sizes with complementary art products such as paints, brushes, or art supplies. Offering these as value-added packages can increase the average order value, improve customer experience, and generate additional revenue opportunities

Example:

Instead of:

Canvas = \$100

Sell:

Premium Canvas + Paint Set + Brushes = \$140

Customer feels: "I am getting a complete package."

Business gets: Higher revenue per transaction.

6) Delete duplicate records from work, product_size, subject and image_link tables

Insight / Finding:

The analysis identifies duplicate records present across the dataset tables. Duplicate entries can create redundancy, increase storage usage, and negatively impact the accuracy of analysis by causing incorrect counts, inflated metrics, or inconsistent reporting.

Possible Reasons / Investigation Areas:

Data entry errors:

The same artwork or record may have been entered multiple times manually.

Data import issues:

Duplicates may occur while merging data from multiple sources or during repeated data uploads.

Lack of unique identifiers:

Tables may not have proper primary keys or validation rules to prevent repeated records.

System synchronization issues:

Data duplication may happen when different systems update the same records without proper controls.

Business Impact / Recommendation:

Maintaining clean and accurate data is essential for reliable decision-making. Duplicate records should be removed and preventive checks should be implemented, such as unique constraints, validation rules, and regular data quality audits. This ensures accurate reporting, better inventory management, and trustworthy business insights.

7) Identify the museums with invalid city information in the given dataset

Insight / Finding:

The analysis identifies museums with invalid city information, such as incorrect formats, numerical values, or inconsistent entries in the city field. These records indicate data quality issues that may affect location-based analysis and reporting accuracy.

Possible Reasons / Investigation Areas:

Data entry errors:

City names may have been entered incorrectly due to manual mistakes.

Incorrect data format:

Some city fields may contain numbers, special characters, or inconsistent formatting instead of valid city names.

Data migration issues:

Errors may occur when transferring data from external systems or combining multiple datasets.

Missing validation rules:

The system may not have proper checks to prevent invalid location information from being entered.

Incomplete or corrupted records:

Some museum location details may have been damaged or incorrectly stored.

Business Impact / Recommendation:

Invalid city information can impact geographical analysis, museum location reporting, and decision-making. The organization should validate and standardize location data, implement data entry rules, and perform regular data quality checks to ensure accurate reporting and reliable insights.

8) The Museum_Hours table has 1 invalid entry. Identify it and remove it.

Insight / Finding:

The analysis identifies an invalid record present in the Museum_Hours table. Incorrect museum operating hour information can create inconsistencies in reporting and may affect visitor planning, operational analysis, and customer experience.

Possible Reasons / Investigation Areas:

Data entry mistake:

The incorrect hour value may have been entered manually due to human error.

Invalid time format:

The record may contain incorrect time values, inconsistent formats, or impossible timings.

Data migration issue:

The invalid entry may have been introduced while importing or transferring data from another system.

Missing validation rules:

The system may not have restrictions to prevent incorrect operating hours from being recorded.

Outdated information:

Museum timings may have changed, but the old record was not updated properly.

Business Impact / Recommendation:

Incorrect museum operating hours can mislead visitors and affect operational reporting. The invalid record should be corrected or removed after verification, and validation rules should be implemented to ensure future entries follow proper time formats and maintain accurate museum information.

9) Fetch the top 10 most famous painting subjects**Insight / Finding:**

The analysis identifies the top 10 most popular painting subjects based on their frequency in the dataset. These subjects represent artworks with higher visibility and potential customer interest, which can help museums and art sellers understand popular themes and optimize their collection strategy.

Possible Reasons / Investigation Areas:**High public interest:**

Certain subjects may attract more attention from visitors and art collectors due to their popularity or emotional connection.

Artist reputation:

Paintings related to famous artists or well-known artistic styles may have higher recognition and demand.

Historical or cultural importance:

Some subjects may be more popular because of their historical significance or cultural value.

Market trends:

Subjects that are currently popular among collectors, online communities, or social media platforms may experience increased demand.

Exhibition impact:

Subjects displayed in famous museums or major exhibitions may gain more recognition and popularity.

Business Impact / Recommendation:

Museums and art sellers can use this information to improve collection planning and marketing strategies. Popular painting subjects can be prioritized for exhibitions, promoted through marketing campaigns, and positioned in high-visitor locations to increase engagement.

For high-demand subjects, businesses can consider:

- Creating premium pricing strategies based on demand and valuation.
- Organizing auctions or special exhibitions.
- Maintaining sufficient inventory or acquiring similar artworks.
- Running promotional campaigns to increase awareness and attract collectors

10) Identify the museums that are open on both Sunday and Monday. Display the museum name and city.

Insight / Finding:

The analysis identifies museums that operate on both Sunday and Monday, providing insights into museum availability and operational scheduling. These museums may have extended operating strategies to capture visitors across both weekend and weekday periods.

Possible Reasons / Investigation Areas:

Higher visitor demand:

Museums may remain open on both days due to consistent visitor traffic and strong customer interest.

Tourist attraction factor:

Museums located in popular tourist destinations may operate on more days to serve visitors with different schedules.

Revenue optimization strategy:

Opening on additional days allows museums to increase ticket sales, merchandise revenue, and overall visitor engagement.

Target audience availability:

Some museums may keep Monday operations because visitors such as tourists, students, or flexible working groups may prefer weekday visits.

Special exhibitions or events:

Museums may remain open on Monday due to ongoing exhibitions, events, or increased demand for specific collections.

Business Impact / Recommendation:

Museums operating on both Sunday and Monday can be analyzed further to compare visitor patterns, revenue, and engagement between weekdays and weekends.

Based on visitor behavior, museums can:

- Introduce targeted promotions or discounts on lower-footfall days.
- Promote less-viewed artworks during slower periods.
- Schedule special exhibitions or events on high-demand days.
- Optimize staffing and resource allocation based on visitor trends.

11) How many museums are open every single day?

Insight / Finding:

The analysis identifies museums that operate throughout the entire week, indicating museums with maximum visitor accessibility. These museums may have strategic operating schedules designed to maximize visitor reach, engagement, and revenue opportunities.

Possible Reasons / Investigation Areas:

High visitor demand:

Popular museums may remain open every day to accommodate consistent visitor traffic and maximize ticket revenue.

Tourist destination factor:

Museums located in tourist-heavy areas may operate daily because visitors arrive throughout the week.

Revenue generation strategy:

Daily operations allow museums to increase revenue opportunities through ticket sales, exhibitions, merchandise, and additional services.

Low visitor demand possibility:

Some museums may operate every day despite low footfall because they are trying to increase visibility, attract visitors, or maintain regular operations.

Location impact:

Museums located in less crowded areas may need additional marketing efforts to attract more visitors.

Operational decision:

Some museums may remain open daily due to special exhibitions, events, or institutional policies.

Business Impact / Recommendation:

Museums open every day should be analyzed further by comparing operating days with visitor count, ticket revenue, and engagement metrics to understand whether daily operations are generating value.

For high-performing museums:

- Introduce premium experiences such as exclusive tours or special access tickets for rare collections.
- Organize special exhibitions or events to increase revenue.
- Add additional services such as cafes, souvenir shops, or visitor experiences to increase average customer spending.

For low-performing museums:

- Improve marketing campaigns and digital presence.
- Introduce loyalty programs or repeat visitor benefits.
- Analyze location accessibility and customer preferences to increase footfall.

12) Which are the top 5 most popular museums? (Popularity is defined based on the most no of paintings in a museum)

Insight / Finding:

The analysis identifies the top 5 museums with the highest number of paintings in their collection. These museums represent locations with larger art inventories and potentially greater cultural significance due to the volume of artworks available.

Possible Reasons / Investigation Areas:

Large art collection:

These museums may have accumulated a larger number of paintings over time through acquisitions, donations, or historical collections.

Historical importance:

Museums with a long history may naturally have larger collections compared to newer museums.

Strong artist relationships or acquisitions:

Some museums may have better access to artwork from renowned artists or collectors.

Higher funding and resources:

Well-funded museums may have greater capability to acquire, preserve, and display more artworks.

Collection strategy:

Some museums may specialize in specific art categories, leading to a higher number of paintings.

Business Impact / Recommendation:

Museums with larger collections can be leveraged as major attraction points by improving visitor engagement and maximizing the value of their artwork inventory.

Possible strategies:

- Promote these museums through digital marketing and online campaigns to increase awareness.
- Organize special exhibitions featuring rare or unique artworks to attract visitors.
- Introduce premium experiences such as guided tours, exclusive access, or special exhibitions.
- Use high-traffic museums for displaying valuable and popular artworks.
- Introduce additional revenue streams such as cafes, merchandise stores, memberships, or workshops.

13) Who are the top 5 most popular artists? (Popularity is defined based on the highest number of paintings created by an artist)

Insight / Finding:

The analysis identifies the top 5 artists with the highest number of paintings in the dataset. These artists represent creators with a significant volume of artwork and strong representation within the museum collection.

Possible Reasons / Investigation Areas:

Long artistic career:

Artists with longer careers may have produced a larger number of paintings over time.

High productivity:

Some artists may have created a large number of artworks due to their working style, creativity, or artistic approach.

Historical importance:

Artists with significant cultural or historical influence may have accumulated larger collections in museums.

Museum acquisition strategy:

Museums may intentionally collect more artworks from certain artists due to their importance or relevance.

High quantity but lower demand possibility:

A large number of paintings does not always indicate high market popularity. Some artists may have many artworks but lower customer interest or valuation.

Inventory management concern:

If many artworks from an artist remain unsold or undisplayed for a long time, it may indicate lower market demand or limited audience interest.

Business Impact / Recommendation:

The museums and art organizations can use this analysis to understand which artists have strong representation in their collections and optimize exhibition strategies.

Possible actions:

- Organize special exhibitions or artist-focused events to attract visitors interested in those artists.
- Highlight selected artworks from these artists in high-visibility museum areas.
- Analyze visitor engagement and sales performance to determine actual demand.
- Promote lesser-known artists if their artworks have potential but lack awareness.
- Review inventory performance to identify artists whose paintings require different marketing strategies.

-- 14) Display the 3 least popular canva sizes

The analysis identifies the 3 least frequently used canvas sizes in the dataset. These canvas sizes have lower representation compared to other sizes, which may indicate lower artist preference, limited demand, or potential inventory utilization issues.

Possible Reasons / Investigation Areas:

Limited artist preference:

Artists may prefer other canvas dimensions that better suit common painting styles or artwork requirements.

Better alternatives are available:

Other canvas sizes may provide better value, pricing, or usability, reducing demand for these sizes.

Quality concerns:

Lower-quality material, durability issues, or poor performance may discourage artists from choosing these canvas sizes.

Pricing factors:

The price may not be competitive compared to similar canvas sizes available in the market.

Marketing and awareness issues:

Artists may not be aware of these canvas sizes due to limited promotion or visibility.

Specific use-case limitation:

Some canvas sizes may only be suitable for specific artwork types, reducing overall demand.

Business Impact / Recommendation:

The business should analyze why these canvas sizes have lower demand by comparing them with high-performing canvas sizes in terms of price, quality, sales volume, and customer preferences.

Possible strategies:

- Bundle less popular canvas sizes with best-selling art supplies to increase adoption.
- Offer promotional discounts or combo packages to clear slow-moving inventory.
- Collect feedback from artists to understand usability issues.
- Improve marketing efforts if the issue is lack of awareness.
- Reassess pricing if similar canvas sizes are available at better prices.

15) Which museum has the highest number of paintings from the most popular painting style?

Insight / Finding:

The analysis identifies the museum that has the highest representation of paintings from the most popular painting style in the dataset. This helps understand which museums have stronger collections around highly preferred artistic styles.

Possible Reasons / Investigation Areas:

Strong collection strategy:

The museum may have intentionally acquired more artworks from this style due to its popularity and cultural importance.

Historical or artistic importance:

Certain painting styles may have higher recognition because of their influence, uniqueness, or connection with famous artists.

Higher audience interest:

Popular painting styles may attract more attention from visitors, collectors, and art enthusiasts.

Museum specialization:

The museum may focus on a particular artistic movement or style, resulting in a larger collection.

Acquisition opportunities:

The museum may have better access to artists or collections related to that style.

Business Impact / Recommendation:

Museums with strong collections in popular painting styles can leverage this advantage to improve visitor engagement and revenue opportunities.

Possible strategies:

- Organize special exhibitions focused on the popular painting style.
- Introduce workshops or art-learning sessions related to that style.
- Invite artists who specialize in the same style for live demonstrations or events.
- Promote these collections through digital marketing and social media campaigns.
- Acquire more artworks from this style to strengthen the museum's collection.
- Place these popular artworks in high-visibility areas to improve visitor experience.

16) Identify the artists whose paintings are displayed in multiple countries

Insight / Finding:

The analysis identifies artists whose artworks are present across multiple countries, indicating artists with broader international representation and global reach within the museum network.

Possible Reasons / Investigation Areas:

International recognition:

Artists may have gained global attention due to their unique style, artistic contribution, or historical importance.

High cultural significance:

Some artists may represent important art movements or cultural heritage, leading museums worldwide to collect their works.

Strong collector demand:

Artists with high demand among collectors may have their artworks acquired by museums in different regions.

Museum acquisition strategy:

Museums across countries may intentionally collect artworks from internationally recognized artists to attract visitors.

Artist reputation and legacy:

Established artists may have wider representation due to their long-term influence in the art world.

Business Impact / Recommendation:

Artists with international representation can be considered valuable assets for museum engagement and collection planning.

Possible strategies:

- Promote exhibitions featuring internationally recognized artists.
- Highlight global artists through marketing campaigns to attract art enthusiasts.
- Organize special events, workshops, or artist-focused exhibitions.
- Analyze visitor interest and artwork performance before acquiring more paintings.
- Strengthen international partnerships and collaborations with museums.

17) Display the country and city with the most museums

Insight / Finding:

The analysis identifies the country and city that have the highest concentration of museums. This helps understand regions with strong art infrastructure, cultural importance, and higher representation of museums.

Possible Reasons / Investigation Areas:**Historical importance:**

Certain cities may have a rich artistic history, leading to more museums being established.

Tourism factor:

Popular tourist destinations may have more museums to attract visitors.

Cultural investment:

Some regions may invest more in preserving art, heritage, and cultural assets.

Economic development:

Developed cities may have better resources to establish and maintain museums.

Population density:

Large cities may have higher demand for cultural and educational attractions.

Business Impact / Recommendation:

Cities with a high concentration of museums can be analyzed for tourism and visitor opportunities. Museums in these regions can collaborate, organize cultural events, and promote art experiences to increase visitor engagement.

Possible strategies:

- Create museum collaboration programs.
- Promote art tourism packages.
- Analyze visitor trends to improve exhibitions.
- Invest in marketing for high-potential locations.

18) Identify the artist and museum where the most expensive and least expensive painting is placed**Insight / Finding:**

The analysis identifies the highest and lowest priced paintings along with their artists, museums, locations, and canvas details. This helps understand premium artworks as well as lower-valued assets within the collection.

Possible Reasons / Investigation Areas:

High-priced painting reasons:

- Famous artist reputation.
- Historical or cultural importance.
- Rare artwork availability.
- High collector demand.

Low-priced painting reasons:

- Lesser-known artist.
- Lower market demand.
- Condition issues.
- Limited recognition.

Museum placement factors:

- Premium artworks may be displayed in high-profile museums.
- Lower-valued artworks may have limited visibility.

Business Impact / Recommendation:

High-value paintings can be prioritized for premium exhibitions, insurance evaluation, and marketing campaigns. Lower-valued artworks should be reviewed to understand whether they need better promotion, repositioning, or valuation updates.

19) Which country has the 5th highest number of paintings?

Insight / Finding:

The analysis identifies the country ranking fifth based on the total number of paintings represented in the dataset. This highlights countries with significant contribution to the global art collection.

Possible Reasons / Investigation Areas:

Strong artistic heritage:

Countries with rich art history may contribute more artworks.

Number of active artists:

Some countries may have larger artistic communities.

Museum presence:

Countries with more museums may naturally have more recorded paintings.

Cultural preservation:

Some countries may invest heavily in maintaining art collections.

Business Impact / Recommendation:

The identified country can be analyzed for potential partnerships, exhibitions, and international art collaborations. Understanding country-wise representation can help museums diversify collections and target global audiences.

20) Which are the 3 most popular and 3 least popular painting styles?**Insight / Finding:**

The analysis identifies the most and least represented painting styles based on the number of artworks. This helps understand artistic preferences and collection trends.

Possible Reasons / Investigation Areas:**Popular styles:**

- Higher audience demand.
- Historical importance.
- Greater artist contribution.
- Strong cultural influence.

Less popular styles:

- Limited awareness.
- Changing market preferences.
- Niche audience.
- Lower representation in collections.

Business Impact / Recommendation:

Popular styles can be highlighted through exhibitions, promotions, and educational events. Less popular styles should be analyzed to understand whether they require better marketing, improved placement, or collection strategy changes.

21) Which artist has the most Portrait paintings outside USA?**Insight / Finding:**

The analysis identifies the artist with the highest number of portrait paintings displayed outside the USA. This highlights artists with strong international representation in portrait artwork.

Possible Reasons / Investigation Areas:**International recognition:**

The artist may have strong global demand and reputation.

Portrait popularity:

Portrait artworks may have higher appreciation among collectors and museums.

Cultural significance:

The artist may represent an important artistic movement or era.

Museum acquisition trends:

International museums may prefer collecting works from globally recognized artists.

Business Impact / Recommendation:

Artists with strong international portrait representation can be leveraged for global exhibitions, marketing campaigns, and collection expansion. Museums can analyze visitor interest and consider collaborations around these artists.